

Software Architecture for Big Data - Exercise 6

This exercise we are going to add an object storage to our setup. We are going to use [Minio](#) as an S3 compatible local object storage provider. To simplify things, we are going to call this object store S3.

For every posted drink order we want to create a "receipt" markdown file, in addition to storing the order in the database. The receipt file should conform to the following format:

```
# Order: 11
```

Created At	Drink ID	Amount	
-----	-----	-----	
Nov 12 17:12:39	1	3	

```
Thanks for drinking with us!
```

Todo

- ☐ Add a Minio instance to the existing `docker-compose.yml`
 - Expose port 8500
 - Persist storage using a volume
 - Set the correct environment variables
- ☐ Create a markdown compatible receipt string in `model/order.go`
 - Make use of `fmt.Sprintf` to fill the string with data
- ☐ Add Put and Get receipts functionality to the `rest/api.go` file
 - Add `PutObject` to `PostOrder` route
 - Add `GetObject` to `GetReceiptFile` route and serve the requested receipt markdown file
- ☐ Verify that files are stored properly using an S3 compatible viewer

Tips and Tricks

Use the `debug.env` for development, **NOT** in your `docker-compose.yml`.

To rebuild the Orderservice Dockerfile when using Docker Compose, use the following command:

```
docker compose build orderservice
docker compose up -d --force-recreate orderservice
```

Beware that the dashboard will no be served when developing locally! Only the OpenAPI definition is reachable at `http://localhost:3000/openapi/index.html`.

Object Storage (Minio / S3)

To view files in your local S3 / object storage in your IDE, install the [Remote File Systems](#) plugin. Use the username and password supplied in `debug.env` or your `docker-compose.yml`.

Tools > Big Data Tools

Storages

- Orderservice S3

☒ Enable connection ☐ Per project

Name: Orderservice S3

Endpoint: localhost:8500

☐ Custom roots ☒ All buckets in the account ?

Buckets filter: All buckets are visible Filter type: Contain

Access key: root

Secret key:

HTTP proxy: Disabled

Test connection

> Extended Connection Settings

OK Cancel Apply

A [similar plugin](#) also exists for Visual Studio Code.